

Questions with Answer Keys

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Q1 - Single Correct

From the point P , perpendicular tangents PQ and PR are drawn to ellipse $x^2 + 4y^2 = 4$. Locus of circumcentre of ΔPQR is

(1) $x^2 + y^2 = \frac{16}{5}(x^2 + 4y^2)^2$

(2) $x^2 + y^2 = \frac{5}{16}(x^2 + 4y^2)^2$

(3) $x^2 + 4y^2 = \frac{16}{5}(x^2 + y^2)^2$

(4) $x^2 + 4y^2 = \frac{5}{16}(x^2 + y^2)^2$

Q2 - Single Correct

The normal to the ellipse $4x^2 + 5y^2 = 20$ at the point P touches $y^2 = 4x$. The eccentricity angle of point P is

(1) $\frac{\pi}{2} + \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$

(2) $\frac{\pi}{2} + \tan^{-1}\left(\frac{1}{\sqrt{5}}\right)$

(3) $\pi - \tan^{-1}\left(\frac{1}{\sqrt{5}}\right)$

(4) $\pi - \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$

Q3 - Single Correct

An ellipse with major axis and minor axis of length $10\sqrt{3}$ and 10 respectively, slides along the coordinate axes and always remains confined in the first quadrant. The locus of centre of the ellipse will be the arc of a circle.

The length of this arc will be equal to

(1) 10π

(2) 5π

(3) $\frac{5\pi}{4}$

(4) $\frac{5\pi}{3}$

Q4 - Single Correct

Point O is the centre of ellipse with major axis AB and minor axis CD . Point F is one focus of the ellipse. If $OF = 6$ and the diameter of inscribed circle of ΔOCF is 2, then the product $(AB)(CD)$ is equal to

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(1) 65

(2) 52

(3) 78

(4) None of these

Q5 - Single Correct

Let P be a point on ellipse $9x^2 + 16y^2 = 144$ whose foci are S, S' . If A, B are feet of perpendiculars from S, S' respectively to the tangent at P , and if normal at P passes through S meet the ellipse at Q and O is origin, then $\frac{SP+SQ}{SP \cdot SQ} + \frac{OA \cdot OB}{SA \cdot SB}$ is

(1) $9/24$ (2) $25/9$ (3) $24/9$ (4) $9/16$

Q6 - Single Correct

For ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, four normals cannot be drawn through a point $(h, 0)$ unless

(1) $|h| > ae$ (2) $|h| \leq ae^2$ (3) $|h| \leq ae$

(4) None of these

Q7 - Single Correct

A circle C intersecting the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$. Such that common chord is of maximum length and intersecting the circle $x^2 + y^2 - 6x - 10y + 9 = 0$ orthogonally, then the number of common tangents of circle C and circle $x^2 + y^2 - 10x - 11 = 0$ is

(1) 0

(2) 2

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(3) 3

(4) 4

Q8 - Single Correct

If semi-major axis of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 (a > b)$ is 20 units and $\sum_{i=1}^{20} (S_1 P_i) (S_2 Q_i) = 5120$, where P_i and Q_i denotes feet of perpendicular drawn from foci S_1 and S_2 on a tangent $T_i = 0$, then the eccentricity of ellipse is

(1) $\frac{1}{5}$ (2) $\frac{\sqrt{89}}{10}$ (3) $\frac{3}{5}$ (4) $\frac{\sqrt{7}}{8}$

Q9 - Single Correct

Let the major axis of a standard ellipse equals the transverse axis of a standard hyperbola and their director circles have radius equal to $2R$ and R , respectively. If e_1 and e_2 are the eccentricity of the ellipse and hyperbola, then the correct relation is

(1) $4e_1^2 - e_2^2 = 6$ (2) $e_1^2 - 4e_2^2 = 6$ (3) $4e_2^2 - e_1^2 = 6$

(4) None of these

Q10 - Single Correct

For each positive integer n , consider the point P with abscissa n on the curve $y^2 - x^2 = 1$. If d_n represents the shortest distance from the point P to the line $y = x$. Then, $\lim_{n \rightarrow \infty} (n \cdot d_n)$ has the value equal to

(1) $\frac{1}{2\sqrt{2}}$ (2) $\frac{1}{2}$ (3) $\frac{1}{\sqrt{2}}$

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(4) 0

Q11 - Single Correct

If a ray of light incident along the line $3x + (5 - 4\sqrt{2})y = 15$ gets reflected from the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$, then its reflected ray goes along the line

(1) $\sqrt{2}x - y + 5 = 0$

(2) $\sqrt{2}y - x + 5 = 0$

(3) $\sqrt{2}y - x - 5 = 0$

(4) $3x - y(4\sqrt{2} + 5) + 15 = 0$

Q12 - Single Correct

A conic passes through the point (2,4) and is such that the segment of any of its tangents at any point contained between the coordinate axes is bisected at the point of tangency. Then, the foci of conic are

(1) $(2\sqrt{2}, 0)$ and $(-2\sqrt{2}, 0)$

(2) $(2\sqrt{2}, 2\sqrt{2})$ and $(-2\sqrt{2}, -2\sqrt{2})$

(3) (4,4) and (-4,-4)

(4) $(4\sqrt{2}, 4\sqrt{2})$ and $(-4\sqrt{2}, -4\sqrt{2})$

Q13 - Single Correct

A line is such that its intercepts on the coordinate axes are e and e' , where e and e' are the eccentricity of a hyperbola and its conjugate hyperbola. If this line is touching a fixed circle, then the area of circle is

(1) $\pi/2$

(2) π

(3) $3\pi/2$

(4) 2π

Q14 - Single Correct

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$P(x, y)$ satisfy $x^2 + y^2 = 1$, let maximum value of $(x + y)^2$ is λ , then the number of tangents drawn from the point $(\lambda, 0)$ to hyperbola $(x - 2)^2 - y^2 = 1$ is

- (1) 0
- (2) 1
- (3) 2
- (4) 4

Q15 - Multiple Correct

The tangent at any point P on a standard ellipse with foci S and S' meets the tangents at the vertices A and A' in the points V and V' , then

- (1) $l(AV) \cdot l(A'V') = b$
- (2) $l(AV) \cdot l(A'V') = b^2$
- (3) $\angle V'SV = 90^\circ$
- (4) $V'S'VS$ is cyclic quadrilateral

Q16 - Multiple Correct

Extremities of latusrectum of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$) having a major axis $2a$ lies on

- (1) $x^2 = a(a - y)$
- (2) $x^2 = a(a + y)$
- (3) $y^2 = a(a + x)$
- (4) $y^2 = a(a - x)$

Q17 - Multiple Correct

A point P moves on XY -plane such that $PS + PS' = 4$, where $S(k, 0)$ and $S'(-k, 0)$, then the locus of P is

- (1) ellipse, if $k \in (-2, 2)$
- (2) pair of coincident lines, if $k = \pm 2$
- (3) empty, if $k \in (-\infty, -2) \cup (2, \infty)$

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(4) None of the above

Q18 - Multiple Correct

If P is a point of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, whose foci are S and S' . Let $\angle PSS' = \alpha$ and $\angle PS'S = \beta$, then

(1) $PS + PS' = 2a$, if $a > b$

(2) $PS + PS' = 2b$, if $b < a$

(3) $\tan \frac{\alpha}{2} \cdot \tan \frac{\beta}{2} = \frac{1-e}{1+e}$

(4) $\tan \frac{\alpha}{2} \cdot \tan \frac{\beta}{2} = \frac{\sqrt{a^2-b^2}}{b^2} \left(a - \sqrt{a^2-b^2} \right)$, when $a > b$

Q19 - Multiple Correct

For the ellipse $x^2 + 2y^2 = 1$, the set of rational points on it can be of the form

(1) $\left(\frac{p^2+q^2}{2p^2+q^2}, \frac{2pq}{p^2+q^2} \right)$

(2) $\left(\frac{2p^2-q^2}{2p^2+q^2}, \frac{-2pq}{2p^2+q^2} \right)$

(3) $\left(\frac{q^2-2p^2}{2p^2-q^2}, \frac{2pq}{2p^2-q^2} \right)$

(4) $\left(\frac{2p^2-q^2}{2p^2+q^2}, \frac{2pq}{2p^2+q^2} \right)$

Q20 - Multiple Correct

Which of the following equations in parametric form can represent a hyperbola, where t is a parameter?

(1) $x = \frac{a}{2} \left(t + \frac{1}{t} \right)$ and $y = \frac{b}{2} \left(t - \frac{1}{t} \right)$

(2) $\frac{tx}{a} - \frac{y}{b} = 1$ and $\frac{x}{a} + \frac{ty}{b} + t = 0$

(3) $x = e^t + e^{-t}$ and $y = e^t - e^{-t}$

(4) $x^2 - 6 = 2 \cos t$ and $y^2 + 2 = 4 \cos^2 \frac{t}{2}$

Q21 - Multiple Correct

Let P and Q be non-zero real numbers. Then, the equation $(px^2 + qy^2 + r)(4x^2 + 4y^2 - 8x - 4) = 0$ represents

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- (1) two straight lines and a circle, when $r = 0$ and p, q are of the opposite sign
- (2) two circles, when $p = q$ and r is of sign opposite to that of p
- (3) a hyperbola and an ellipse, when p and q are of opposite sign and $r \neq 0$
- (4) a circle and an ellipse, when p and q are unequal but of same sign and sign of r opposite to that of p

Q22 - Multiple Correct

Solution of the differential equation $(1 - x^2) \frac{dy}{dx} + xy = ax$, where $a \in R$, is

- (1) a conic which is an ellipse or a hyperbola with principal axes parallel to coordinates axes
- (2) centre of conic is $(0, a)$
- (3) length of one of the principal axes is 1
- (4) length of one of the principal axes is equal to 2

Q23 - Multiple Correct

If θ is eliminated from the equations $a \sec \theta - x \tan \theta = y$ and $b \sec \theta + y \tan \theta = x$ (a and b are constants), then

- (1) the director circle of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
- (2) auxiliary circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
- (3) director circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
- (4) director circle of the circle $x^2 + y^2 = \frac{a^2 + b^2}{2}$

Q24 - Multiple Correct

The tangent to hyperbola, $x^2 - 3y^2 = 3$ at the point $(\sqrt{3}, 0)$, when associated with two asymptotes constitutes

- (1) isosceles triangle
- (2) an equilateral triangle
- (3) a triangle whose area is $\sqrt{3}$ sq units
- (4) a right isosceles triangle

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Q25 - Multiple Correct

The area of the triangle inscribed in an ellipse bears the ratio $\sqrt{5} : 3$ to the area of triangle formed by joining points on the auxilliary circle corresponding to the vertices of the first triangle, then the eccentricity of the ellipse is

(1) $2/3$

(2) $3/4$

(3) $\sqrt{5}/3$

(4) $3/5$

Q26 - Multiple Correct

If a right angled triangle is inscribed in the rectangular hyperbola $xy = c^2$, then the angle between the tangent at the right angled vertex and the hypotenuse of the triangle is

(1) 0

(2) 45°

(3) 60°

(4) 90°

Q27 - Multiple Correct

The number of normal(s) of a rectangular hyperbola which can touch its conjugate is equal to

(1) 0

(2) 2

(3) 4

(4) 8

Q28 - Paragraph 1

Passage I (For Question 29, 30)

If foci of an ellipse is $(-1,2)$ and $(-2,3)$ and its tangent at a point A is $2x + 3y + 9 = 0$.

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The length of minor axis of the ellipse will be

(1) $\sqrt{27/2}$ units

(2) $\sqrt{14}$ units

(3) $2\sqrt{14}$ units

(4) None of these

Q29 - Paragraph 1

The area of ellipse will be

(1) $3\sqrt{21}\pi$ sq units

(2) $\sqrt{203}\pi$ sq units

(3) $21\sqrt{3}\pi$ sq units

(4) None of these

Q30 - Paragraph 2

Passage II (For Question 31, 32)

Let an ellipse having major axis and minor axis parallel to X -axis and Y -axis respectively. Its two foci S and S' are $(2,1)$, $(4,1)$ and a line $x + y = 9$ is a tangent to this ellipse at point P .

Eccentricity of the ellipse is equal to

(1) $1/\sqrt{12}$

(2) $1/\sqrt{13}$

(3) $1/2$

(4) None of these

Q31 - Paragraph 2

The length of major axis is equal to

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(1) $\sqrt{13}$

(2) $2\sqrt{11}$

(3) $\sqrt{52}$

(4) $2\sqrt{12}$

Q32 - Paragraph 3

Passage III (For Question 33, 34)

To prove geometrical properties of ellipse, we may take its standard equation as $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, we will assume that $a > b > 0$ and foci S, S' lie on positive and negative side of X -axis, respectively.

If the normal at any point P meets major and minor axis at G and G' , respectively and OF is the perpendicular drawn from centre O to this normal. Then, $PF \cdot PG$ must be equal to

(1) b^2

(2) a^2

(3) ab

(4) a^2b^2

Q33 - Paragraph 3

If P is any point on the ellipse and G is same as above, then $S'G/SG$ must be

(1) $\frac{OS'}{OS}$

(2) $\frac{G'S}{OG}$

(3) $\frac{S'P}{SP}$

(4) $\frac{a}{b}$

Q34 - Paragraph 4

Passage IV (For Question 35, 36)

The normal at any point $P(x_1, y_1)$ of curve is a line perpendicular to tangent at the point $P(x_1, y_1)$. In case of

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rectangular hyperbola $xy = c^2$, the equation of normal at $(ct, c/t)$ is $xt^3 - yt - ct^4 + c = 0$. The shortest distance between any two curves always exist along the common normal.

If normal at $(5,3)$ of rectangular hyperbola $xy - y - 2x - 2 = 0$ intersect it again at a point

(1) $(-1,0)$

(2) $(-1,1)$

(3) $(0,-2)$

(4) $(3/4, -14)$

Q35 - Paragraph 4

The shortest distance between the parabola $2y^2 = 2x - 1$, $2x^2 = 2y - 1$ is

(1) $2\sqrt{2}$

(2) $\frac{1}{2\sqrt{2}}$

(3) 4

(4) $\sqrt{\frac{36}{5}}$

Q36 - Paragraph 5

Passage V (For Question 37, 38)

A conic C passes through the point $(2,4)$ and is such that the segment of any of its tangents at any point contained between the coordinate axes is bisected at the point of tangency. Let S denotes circle described on the foci F_1 and F_2 of the conic C as diameter.

Vertex of the conic C is

(1) $(2,2), (-2,-2)$

(2) $(2\sqrt{2}, 2\sqrt{2}), (-2\sqrt{2}, -2\sqrt{2})$

(3) $(4,4), (-4,-4)$

(4) $(\sqrt{2}, \sqrt{2}), (-\sqrt{2}, -\sqrt{2})$

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Q37 - Paragraph 5

Directors circle of the conic is

(1) $x^2 + y^2 = 4$

(2) $x^2 + y^2 = 8$

(3) $x^2 + y^2 = 2$

(4) None of these

Q38 - Integer Type

If the curve $x^2 + 3y^2 = 9$ subtends an obtuse angle at the point $(2\alpha, \alpha)$, then a possible integral value of α^2 is

Q39 - Integer Type

From a point perpendicular tangents are drawn to the ellipse $x^2 + 2y^2 = 2$. The chord of contact touches a circle which is concentric with the given ellipse. Then, the ratio of maximum and minimum area of circle is

Q40 - Integer Type

If the normal at the point $P(\theta)$ to the ellipse $\frac{x^2}{14} + \frac{y^2}{5} = 1$ intersecting again at the point $Q(2\theta)$, then the value of $3 \cos \theta + 2$ is

Q41 - Integer Type

The number of points on the ellipse $\frac{x^2}{50} + \frac{y^2}{20} = 1$ from which pair of perpendicular tangents are drawn to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ is

Q42 - Integer Type

$x - 2y + 4 = 0$ is a common tangent to $y^2 = 4x$ and $\frac{x^2}{4} + \frac{y^2}{b^2} = 1$. Then, the value of b^2 is

Q43 - Integer Type

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If equation of the curves on reflection of the ellipse $\frac{(x-4)^2}{16} + \frac{(y-3)^2}{9} = 1$ about the line $x - y - 2 = 0$ is

$16x^2 + 9y^2 + k_1x - 36y + k_2 = 0$, then the units' digit of $(k_1 + k_2)$ is

Q44 - Integer Type

The equation of asymptotes of $xy + 2x + 4y + 6 = 0$ is $xy + 2x + 4y + c = 0$, then the value of c is

Q45 - Integer Type

If a variable line has its intercepts on the coordinate axes e, e' , where $\frac{e}{2}, \frac{e'}{2}$ are the eccentricities of a hyperbola and its conjugate hyperbola, then the line always touches the circle $x^2 + y^2 = r^2$, then the value of r is

Q46 - Integer Type

A chord AB which bisected at $(1,1)$ is drawn to the hyperbola $7x^2 + 8xy - y^2 - 4 = 0$ with centre C , which intersects its asymptotes E and F . If the equation of circumcircle of $\triangle CEF$ is $x^2 + y^2 - ax - by + c = 0$.

Then, the value of $\frac{23(a-b+c)}{12}$ is

Q47 - Integer Type

The latusrectum subtends a right angle at the centre of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If the eccentricity is e , then the value of $[e]$, where $[.]$ denotes greatest integer function is

Q48 - Integer Type

A lattice point (x, y) is a point in XY -plane both of whose coordinates are integers. The number of such lattice points on hyperbola $2xy - 5x + y = 55$ is λ . Then, the value of $\frac{\lambda}{4}$ is

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Answer Key					
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Q1 (2)	Q2 (4)	Q3 (4)	Q4 (1)		
//. mathongo	//. mathongo	//. mathongo	//. mathongo	//. mathongo	//. mathongo
Q5 (3)	Q6 (2)	Q7 (2)	Q8 (3)		
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Q9 (3)	Q10 (1)	Q11 (4)	Q12 (3)		
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Q13 (2)	Q14 (1)	Q15 (2, 3, 4)	Q16 (1, 2)		
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Q17 (1, 2, 3)	Q18 (1, 2, 3)	Q19 (2, 4)	Q20 (1, 3, 4)		
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Q21 (1, 2, 4)	Q22 (1, 2, 4)	Q23 (3, 4)	Q24 (2, 3)		
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Q25 (1)	Q26 (4)	Q27 (3)	Q28 (3)		
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Q29 (4)	Q30 (2)	Q31 (3)	Q32 (1)		
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Q33 (3)	Q34 (4)	Q35 (2)	Q36 (2)		
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Q37 (4)	Q38 (2)	Q39 (4)	Q40 (0)		
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Q41 (4)	Q42 (3)	Q43 (2)	Q44 (8)		
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Q45 (2)	Q46 (4)	Q47 (1)	Q48 (4)		
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